ALFIDO TECH

RIDING THE WAVES OF TECHNOLOGICAL ADVANCEMENT

CUSTOMER BEHAVIOR ANALYSIS

Data Analytics Internship – Project Report

TASK - 1

Organization: Alfido Tech

Internship: Data Analytics Internship

Project: Customer Behaviour Analysis

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Executive Summary

This project focuses on analysing customer purchasing behaviour using transactional data to identify meaningful patterns, customer segments, and opportunities for improving customer engagement and retention.

The analysis was performed on a dataset containing **250,000 transactions from 49,673 unique customers**, covering the period from January 2020 to September 2023. Python, Pandas, NumPy, Matplotlib, and Seaborn were used for data cleaning, analysis, and visualisation.

The project includes exploratory data analysis, customer segmentation using the RFM (Recency, Frequency, Monetary) model, cohort retention analysis, and examination of churn and product return indicators.

The analysis identified that approximately 40% of customers belonged to the At Risk and Hibernating RFM segments. Additionally, 96.61% of customers made multiple purchases during the observed period. The dataset-provided churn indicator marked 20.01% of customers as churned.

Based on these findings, the project proposes targeted customer re-engagement campaigns, loyalty initiatives, improved return-data collection, and stronger retention monitoring.

The overall objective is to convert customer transaction data into actionable business insights that can support better customer relationship and marketing decisions.

Introduction

Customer behaviour analysis helps businesses understand how customers interact with their products and services. By studying transaction history, purchase frequency, spending patterns, and customer activity, businesses can identify valuable customer groups and areas where customer engagement can be improved.

This project analyses e-commerce transaction data to understand customer purchasing behaviour and identify patterns related to customer segmentation, retention, churn, and product returns.

The analysis uses historical transaction records from January 2020 to September 2023. It applies data cleaning, exploratory data analysis (EDA), RFM-based customer segmentation, and cohort analysis to generate meaningful business insights.

Project Objectives

1. To clean and prepare the customer transaction dataset for analysis.
2. To analyse purchasing patterns across product categories, payment methods, age groups, and gender.
3. To segment customers using the Recency, Frequency, and Monetary (RFM) model.
4. To examine customer retention patterns through cohort analysis.
5. To analyse the dataset-provided churn and product return indicators.
6. To identify important business insights and suggest actionable recommendations.

Dataset Description

The dataset used for this project contains e-commerce customer transaction records covering the period from January 2020 to September 2023.

It consists of **250,000 transaction records** belonging to **49,673 unique customers**. Each record represents a purchase transaction and includes information about the customer, purchase date, product category, price, quantity, payment method, and other customer-related indicators.

Dataset Overview

Parameter	Details
Dataset Type	E-commerce Transaction Data
Total Records	250,000
Unique Customers	49,673
Time Period	January 2020 – September 2023
Product Categories	Books, Clothing, Electronics, Home
Payment Methods	Credit Card, PayPal, Cash, Crypto
Data Format	CSV

Key Dataset Attributes

- **Customer Information:** Customer ID, Customer Name, Customer Age, Gender.
- **Transaction Information:** Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount.
- **Customer Behaviour Indicators:** Returns, Churn.
- **Payment Information:** Payment Method.

Note: The dataset does not specify a currency. The `Total Purchase Amount` field is treated as the recorded purchase amount and is not independently verified as revenue.

Data Cleaning and Preprocessing

Before performing the analysis, the dataset was examined and cleaned to improve data quality and ensure consistency.

The following preprocessing steps were performed:

1. **Dataset Inspection:** Checked the dataset dimensions, column names, data types, and overall structure.
2. **Missing Value Treatment:** Identified missing values in the `Returns` column. Missing entries were retained as an `Unknown` return status instead of assuming that the transactions were not returned.
3. **Duplicate Check:** Checked for duplicate records. No exact duplicate rows were found.
4. **Date Conversion:** Converted the `Purchase Date` column into datetime format to enable time-based analysis.
5. **Redundant Column Removal:** Removed the `Age` column because it contained the same information as `Customer Age`.
6. **Feature Engineering:** Created additional columns such as Purchase Year, Purchase Month, Purchase Month Name, Purchase Day, Purchase Day Name, and Age Group.
7. **Data Validation:** Checked customer information for consistency and examined numerical fields for invalid or non-positive values.
8. **Final Dataset Preparation:** Created a cleaned analytical dataset containing 250,000 records and 20 columns for further analysis.

Data Quality Observation: The `Product Price × Quantity` calculation did not match the recorded `Total Purchase Amount` for most transactions. Therefore, the original recorded amount was retained without replacing it with the calculated value.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand customer purchasing patterns and identify trends across different dimensions of the dataset.

The analysis focused on the following areas:

1. Monthly Purchase Trend

Monthly transaction counts were analysed to understand purchasing activity over time.

- March 2021 recorded the highest number of transactions, with **5,914 purchases**.

- Monthly purchase activity remained broadly stable across most of the observed period.
- September 2023 recorded fewer transactions, but this month contains data only up to September 15, so it is not directly comparable with complete months.

2. Product Category Analysis

Purchases and recorded purchase amounts were compared across four product categories.

Product Category	Transactions	Recorded Purchase Amount
Clothing	75,052	204,532,405
Books	74,912	204,939,601
Electronics	50,185	136,599,467
Home	49,851	135,271,210

Key Finding: Clothing recorded the highest number of transactions, while Books generated the highest recorded purchase amount.

3. Payment Method Analysis

Customer transactions were analysed based on the payment methods used.

Payment Method	Transactions	Recorded Purchase Amount
Credit Card	100,486	274,152,396
PayPal	74,837	204,026,613
Cash	49,894	135,705,551
Crypto	24,783	67,458,123

Key Findings:

- Credit Card was the most frequently used payment method.
- PayPal ranked second in transaction volume.
- Crypto recorded the lowest number of transactions.
- Average recorded purchase amounts were similar across all payment methods.

4. Age Group Analysis

Customers were grouped by age to understand differences in purchasing behaviour.

Age Group	Customers	Transactions	Average Recorded Purchase Amount
18–25	7,616	38,349	2,618.18

26–35	9,334	46,906	2,655.71
36–45	9,362	47,186	2,710.81
46–55	9,301	46,753	2,761.48
56–65	9,338	47,120	2,798.70
Above 65	4,722	23,686	2,848.71

Key Findings:

- The 36–45 age group recorded the highest number of transactions.
- The 56–65 age group had the highest total recorded purchase amount.
- Customers above 65 had the highest average recorded purchase amount.
- Average recorded purchase amount generally increased across the age groups.

5. Gender Analysis

Purchasing behaviour was also examined across gender groups.

Gender	Customers	Transactions	Average Recorded Purchase Amount
Female	24,958	125,560	2,727.48
Male	24,715	124,440	2,723.24

Key Finding: Female customers recorded slightly more transactions and a marginally higher average purchase amount than male customers. However, the differences were very small.

Customer Segmentation Using RFM Analysis

RFM analysis is a customer segmentation technique that evaluates customers based on three factors:

- **Recency (R):** How recently a customer made a purchase.
- **Frequency (F):** How often a customer made purchases.
- **Monetary (M):** The total recorded purchase amount associated with a customer.

Each customer was assigned RFM scores from 1 to 5 using relative quintile-based scoring. These scores were then used to classify customers into different segments.

RFM Customer Segments

Customer Segment	Customers	Percentage
Hibernating	11,125	22.40%

At Risk	8,744	17.60%
Champions	8,692	17.50%
Potential Loyalists	7,219	14.53%
Loyal Customers	6,268	12.62%
Others	5,496	11.06%
Recent Customers	2,129	4.29%
Total	**49,673**	**100%**

Key Findings

1. Hibernating customers formed the largest segment, representing 22.40% of all customers.
2. At Risk customers accounted for 17.60% of the customer base.
3. Champions represented 17.50% of customers and can be considered an important group for loyalty initiatives.
4. At Risk and Hibernating customers together accounted for approximately **40% of the customer base**.
5. Recent Customers represented 4.29% of customers.

Business Insight: A considerable proportion of customers belonged to the At Risk and Hibernating segments. Targeted re-engagement campaigns may help encourage these customers to make purchases again.

Note: RFM segments are based on custom scoring rules and relative quintiles. These categories indicate purchasing behaviour, not confirmed customer churn.

Customer Retention and Cohort Analysis

Cohort analysis was performed to understand how customer purchasing activity changed over time. Customers were grouped into cohorts based on the month of their first recorded purchase. Their subsequent monthly purchase activity was then analysed.

Key Findings

- The analysis covered 45 monthly customer cohorts.
- For the January 2020 cohort, 5,376 customers made their first recorded purchase during that month.
- In the second month, 522 customers from this cohort made another purchase, representing **9.71%** of the original cohort.
- In the third month, 578 customers made a purchase, representing **10.75%** of the original cohort.

- Across many subsequent months, established cohorts showed monthly active purchase rates of approximately **9–12%**.

Business Insight

The cohort analysis indicates that a portion of customers continued making purchases in subsequent months. Businesses can use these patterns to identify opportunities for customer engagement through timely reminders, personalised offers, and retention campaigns.

Important Note: The percentages represent customers who made a purchase in a particular month relative to their original cohort size. They do not represent continuous or cumulative customer retention. Later cohorts have shorter observation periods, and September 2023 is an incomplete month.

Churn and Return Analysis

This section examines the dataset-provided customer churn indicator and product return information to understand customer activity and return patterns.

1. Customer Churn Analysis

The churn indicator was analysed at the customer level to understand the proportion of customers marked as churned in the dataset.

Churn Status	Customers	Percentage
Not Churned (0)	39,731	79.99%
Churned (1)	9,942	20.01%
Total	**49,673**	**100%**

Key Findings:

- Approximately 20.01% of customers were marked as churned.
- The remaining 79.99% were marked as not churned.
- Churn percentages across RFM segments were relatively similar, ranging from 19.36% to 20.70%.

Business Insight: Businesses can monitor customer activity and investigate the reasons behind inactivity to develop suitable retention strategies.

Note: The dataset does not specify how the churn indicator was defined or the observation period used. Therefore, it is treated as a dataset-provided indicator rather than independently verified churn.

2. Product Return Analysis

The Returns column was examined to understand returned and non-returned transactions. Missing return values were classified as Unknown.

Return Status	Transactions	Percentage
Not Returned	101,635	40.65%
Returned	100,769	40.31%
Unknown	47,596	19.04%
Total	**250,000**	**100%**

Key Findings:

- 40.31% of all transactions were marked as returned.
- 40.65% were marked as not returned.
- 19.04% had unknown return status.
- Among transactions with known return status, approximately **49.79% were marked as returned**.

Business Insight: The high proportion of returned transactions and missing return information highlights the importance of investigating return reasons and improving return-status data collection.

Key Business Insights

Based on the customer behaviour analysis, the following insights were identified:

1. **Customer Segmentation:** At Risk and Hibernating customers together represented approximately 40% of the total customer base, indicating an opportunity for customer re-engagement.
2. **Repeat Purchasing:** Approximately 96.61% of customers made more than one purchase during the observed dataset period.
3. **Product Category Performance:** Clothing recorded the highest number of transactions (75,052), while Books had the highest recorded purchase amount (204,939,601).
4. **Age-Based Purchasing Patterns:** Average recorded purchase amount generally increased across age groups, with customers above 65 recording the highest average (2,848.71).

5. **Customer Retention:** Cohort analysis showed monthly active purchase rates of approximately 9–12% across many subsequent months for established cohorts.
6. **Churn Indicator:** The dataset marked 20.01% of customers as churned. However, the exact definition of this indicator was not provided.
7. **Product Returns:** Approximately 49.79% of transactions with known return status were marked as returned, highlighting the need to investigate return patterns.

Actionable Business Recommendations

Based on these findings, the following recommendations are proposed:

1. **Re-engagement Campaigns:** Develop targeted email or promotional campaigns for At Risk and Hibernating customers to encourage repeat purchases.
2. **Customer Loyalty Programs:** Introduce exclusive rewards, personalised offers, and loyalty benefits for Champions to strengthen customer relationships.
3. **Age-Based Marketing:** Test personalised campaigns for different age groups based on their purchasing patterns and monitor their conversion rates.
4. **Improve Return Management:** Investigate the reasons behind product returns and improve the collection of missing return-status information.
5. **Retention Monitoring:** Regularly monitor customer cohorts and identify periods of reduced purchasing activity to plan timely engagement campaigns.

Note: These recommendations are proposed actions based on observed patterns. Their actual effectiveness should be evaluated through implementation and performance monitoring.

Conclusion

This project analysed e-commerce customer transaction data to understand purchasing behaviour, customer segments, retention patterns, churn indicators, and product returns.

The analysis of 250,000 transactions from 49,673 unique customers provided useful insights into customer purchasing patterns. RFM segmentation identified that approximately 40% of customers belonged to the At Risk and Hibernating segments. Cohort analysis helped examine monthly customer purchasing activity, while churn and return analysis highlighted areas that require further investigation.

The findings suggest that businesses can use customer transaction data to improve customer engagement, develop targeted marketing strategies, strengthen loyalty initiatives, and monitor retention patterns.

Overall, this project provided practical experience in data cleaning, exploratory data analysis, customer segmentation, data visualisation, and translating analytical findings into actionable business recommendations.

Project Limitations

The following limitations were identified during the analysis:

1. **Purchase Amount Validation:** The recorded Total Purchase Amount did not match Product Price multiplied by Quantity for most transactions. The original recorded values were retained for analysis.
2. **Missing Return Information:** Approximately 19.04% of transactions had unknown return status.
3. **Churn Definition:** The dataset did not provide the exact definition or observation period for the churn indicator.
4. **Retention Measurement:** Cohort analysis measured monthly purchasing activity, not continuous customer retention.
5. **Incomplete Final Month:** September 2023 contained data only up to September 15.
6. **Currency Information:** The dataset did not specify the currency of recorded purchase amounts.

Tools and Technologies Used

Tool / Technology	Purpose
Python	Data analysis and preprocessing
Pandas	Data cleaning and manipulation
NumPy	Numerical operations
Matplotlib	Data visualisation
Seaborn	Statistical visualisation
Google Colab / Jupyter Notebook	Code execution and analysis

Project Deliverables

- Cleaned customer transaction dataset in CSV format.
- Python notebook containing data cleaning, analysis, and visualisations.
- Customer segmentation using RFM analysis.
- Cohort retention analysis.
- Business insights and actionable recommendations.
- Final project report in PDF format.

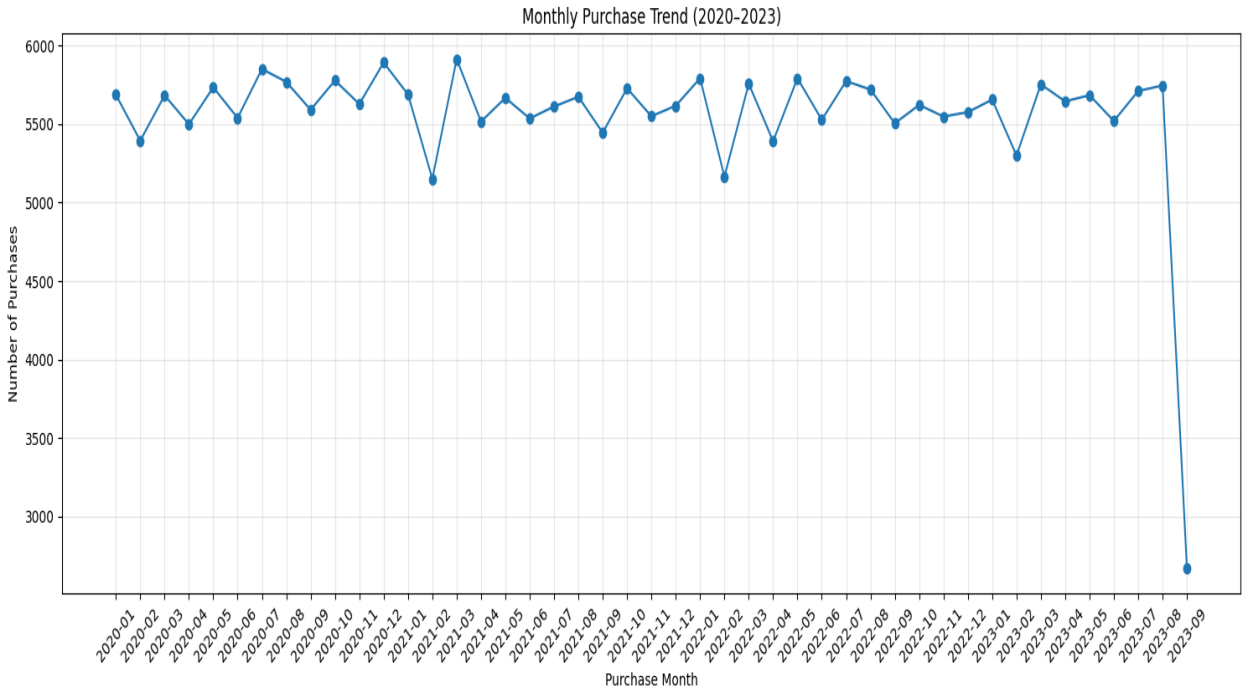


Figure 1: Monthly Purchase Trend (January 2020 – September 2023). September 2023 represents a partial month.

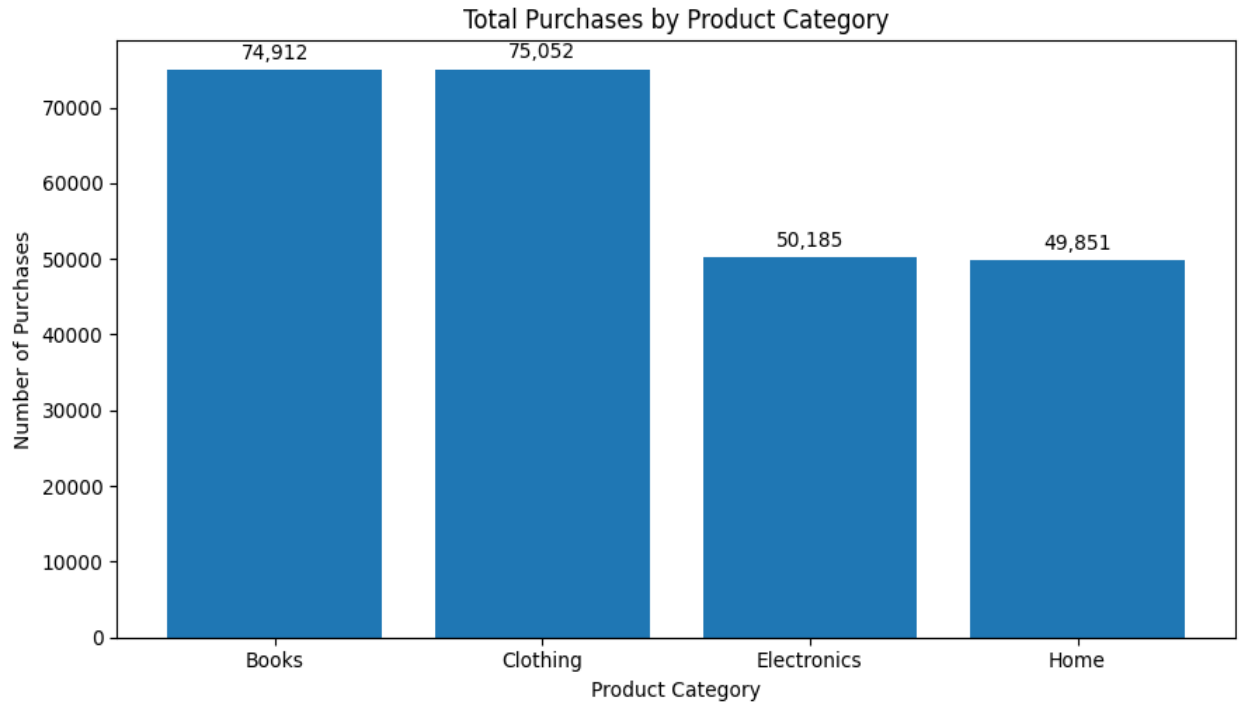


Figure 2: Number of Transactions Across Product Categories.

Observation: Clothing recorded the highest number of transactions, closely followed by Books.

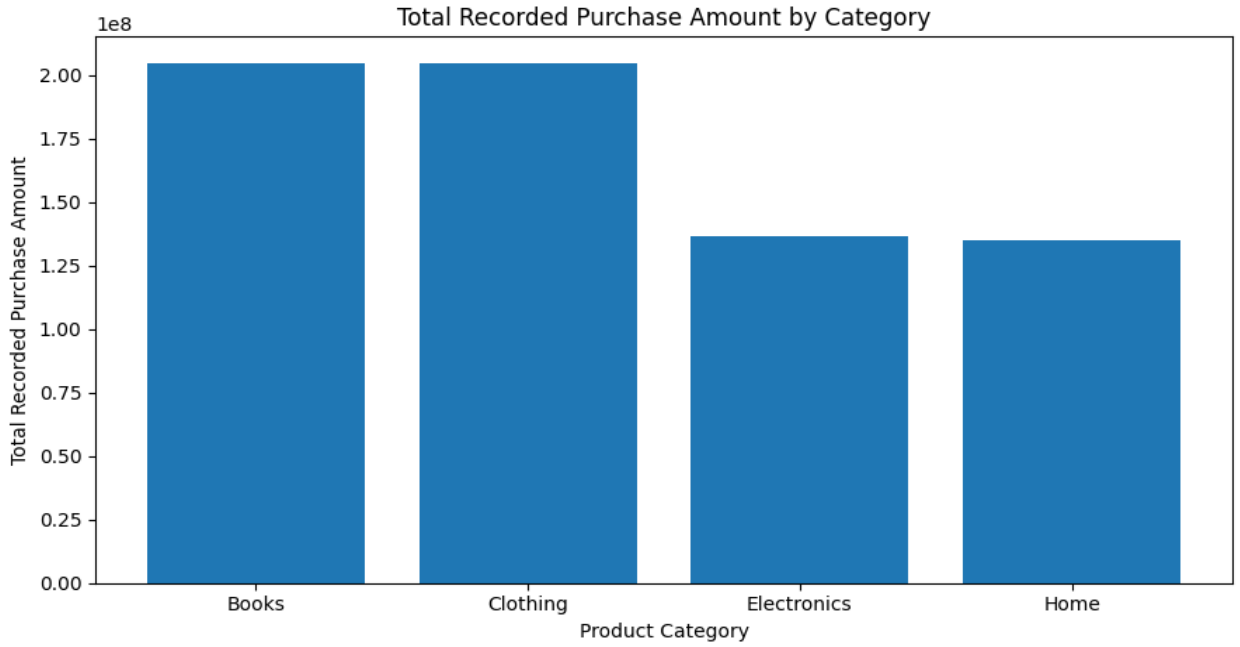


Figure 3: Total Recorded Purchase Amount Across Product Categories

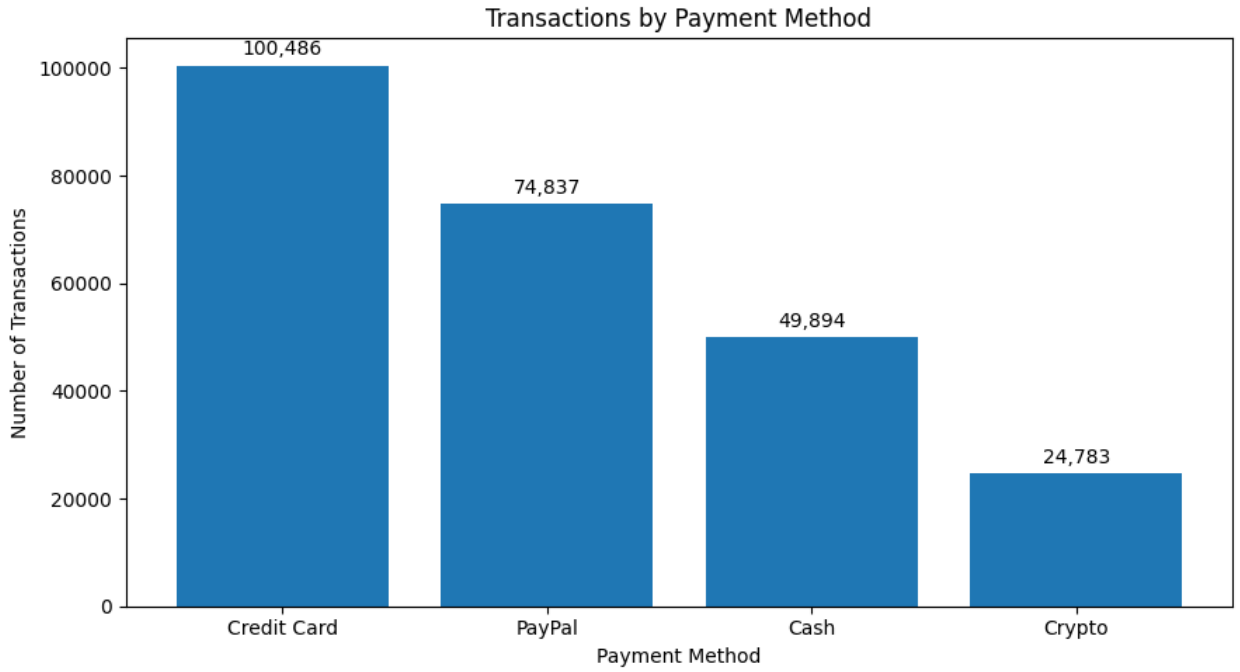


Figure 4: Distribution of Transactions by Payment Method.

Observation: Credit Card was the most frequently used payment method, while Crypto recorded the lowest transaction volume.

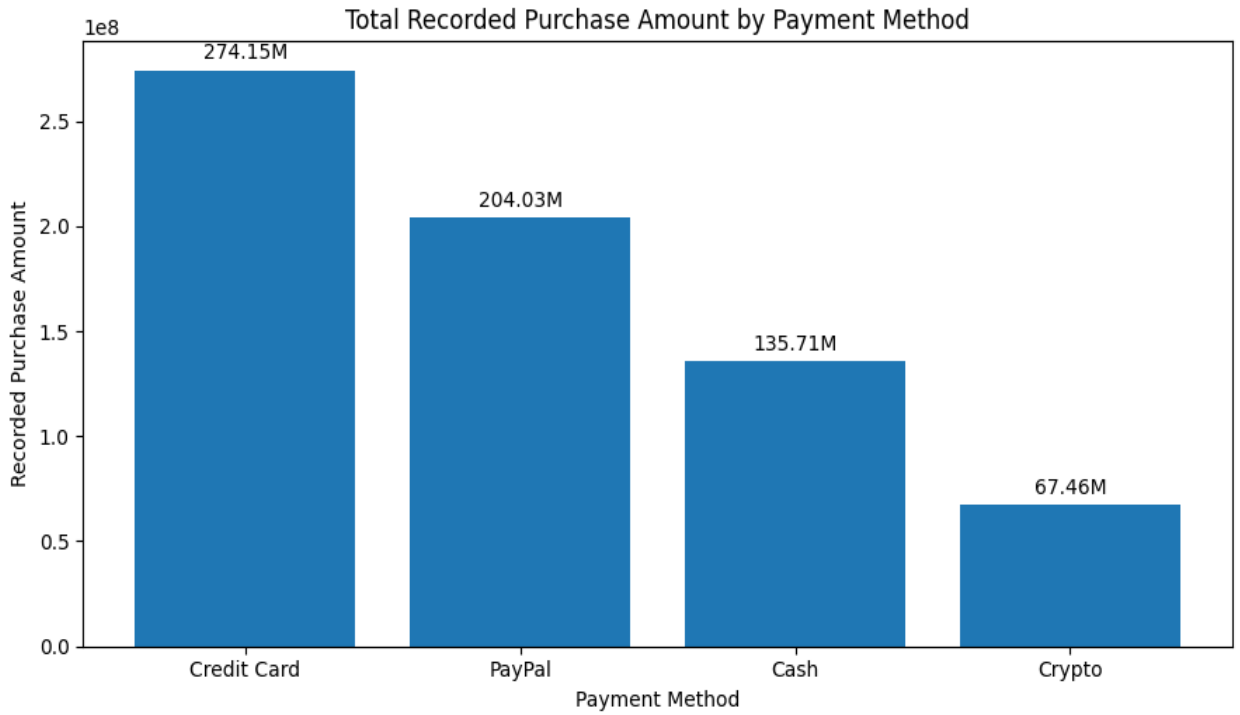


Figure 5: Total Recorded Purchase Amount by Payment Method.

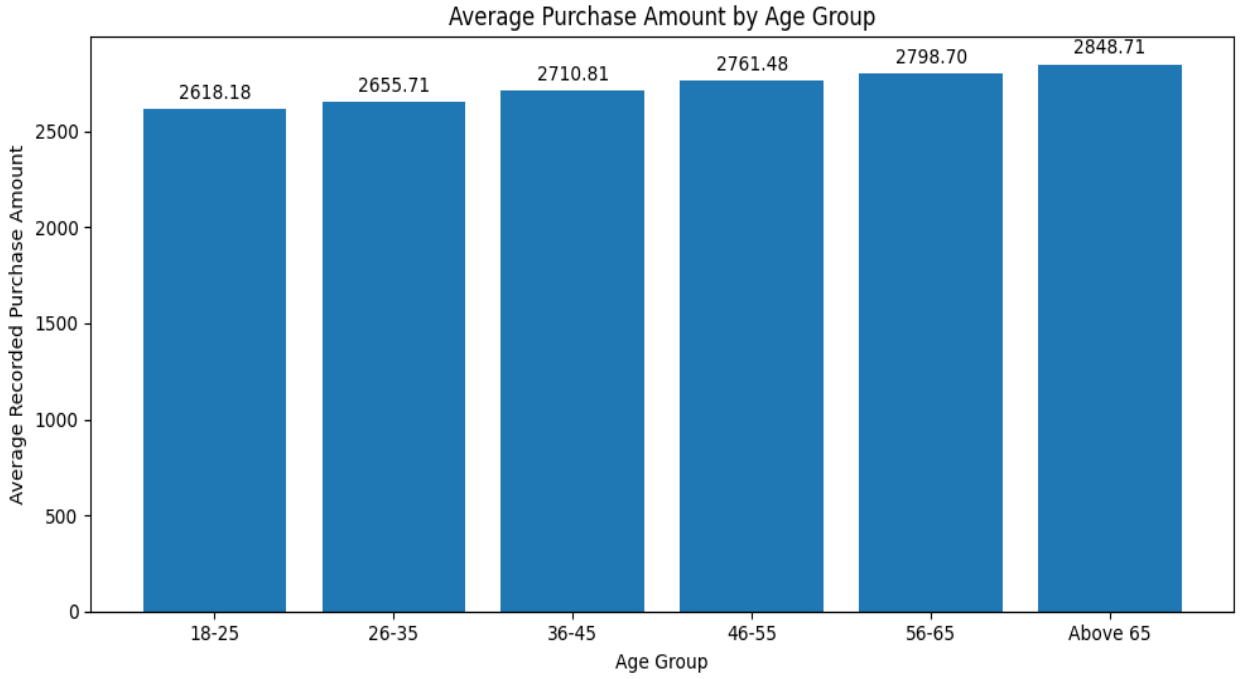


Figure 6: Average Recorded Purchase Amount Across Customer Age Groups.

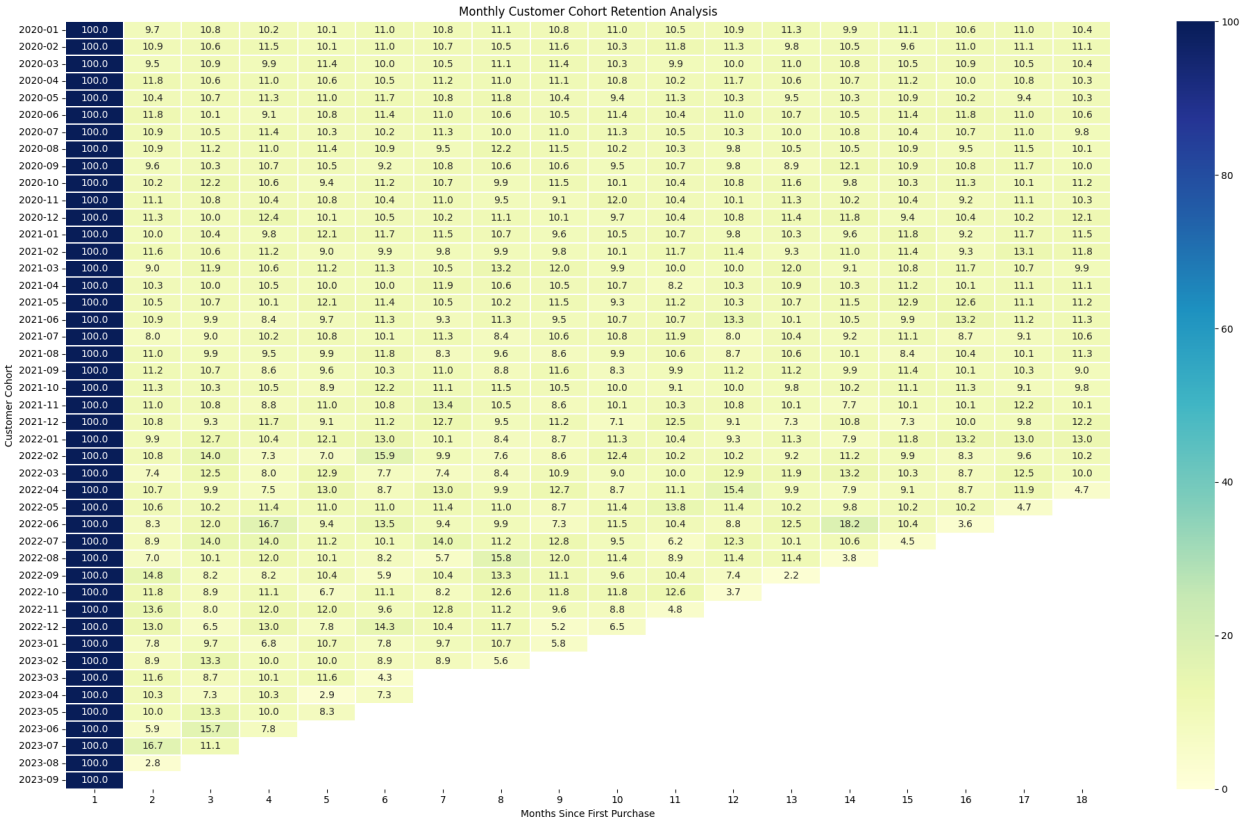


Figure 7: Monthly Customer Cohort Purchase Retention Heatmap.

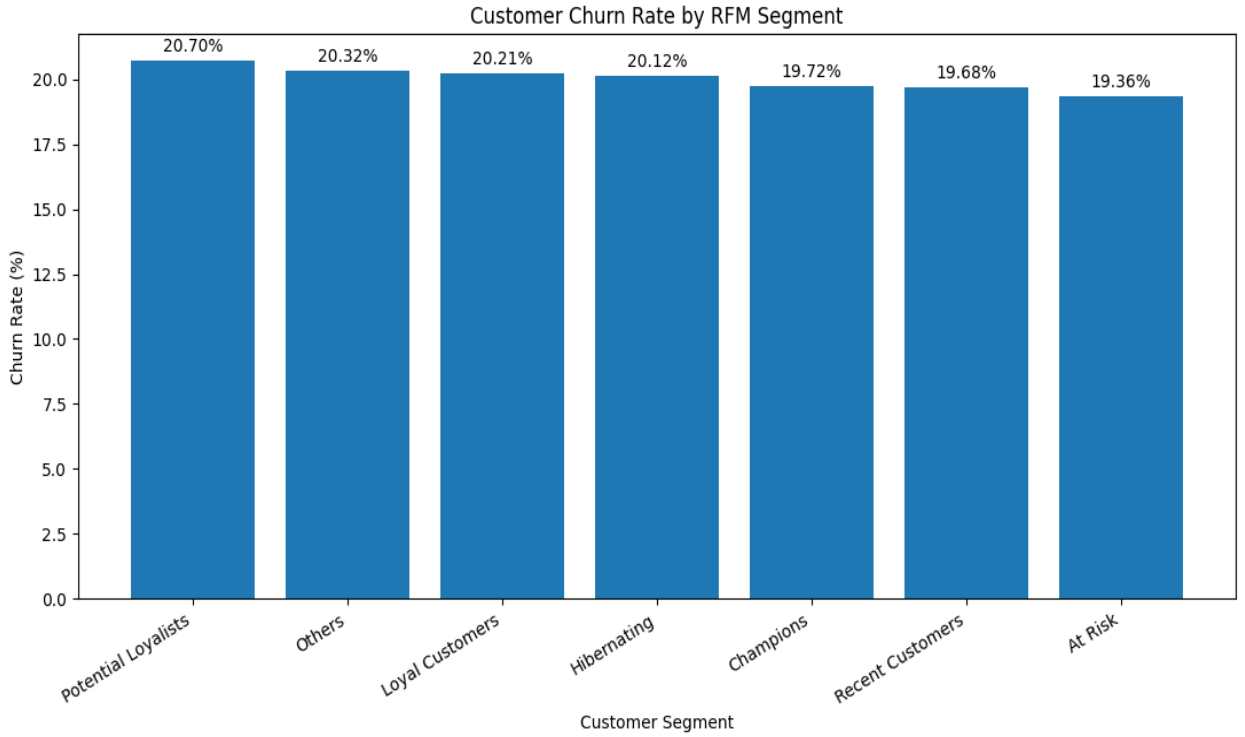


Figure 8: Dataset-Provided Customer Churn Rate Across RFM Segments.

Business Observation:

The churn indicator showed relatively similar rates across RFM segments, ranging from 19.36% to 20.70%. This suggests that the RFM segmentation alone did not strongly differentiate customers based on the dataset-provided churn indicator.